# DeepMine

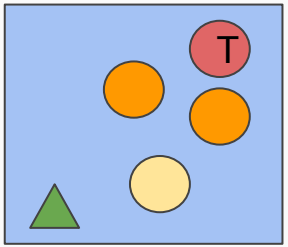
*Ex-Project Pickaxe: Deep Learning Minecraft*

Nikhil Pant - Spencer Greene - Deniz Bayazit

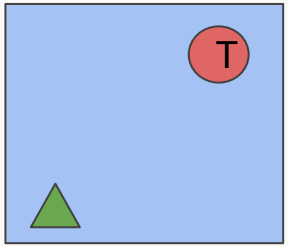
CS 1470 Final Project
Fall 2019
Brown University

# Introduction:
## *Reinforcement Learning Problems*

Dense Reward Environment




Sparse Reward Environment




*Can deep learning overcome the problem of sparse rewards in a particularly complex environment such as the ones presented in the MineRL Competition?*

*This is an important question to ask as sparse rewards are everywhere in the real world and eventually we would like for RL agents to be able to do these tasks in the real world much faster.*

http://minerl.io/index.html

# MineRL Environments

*1) Navigate Dense*

- **Action Space:** attack, back, camera, forward, jump, left, right, sneak, sprint
- **Observation Space:** Screen input, Compass angle
- **Goal:** Navigate to the given goal block

*2) Tree-Chop*

- **Action Space:** Attack, back, camera, forward, jump, left, right, sneak, sprint
- **State Space:** Screen Input
- **Goal:** Collect 64 blocks of wood

# Methodology:
## *Architectures & Baselines*

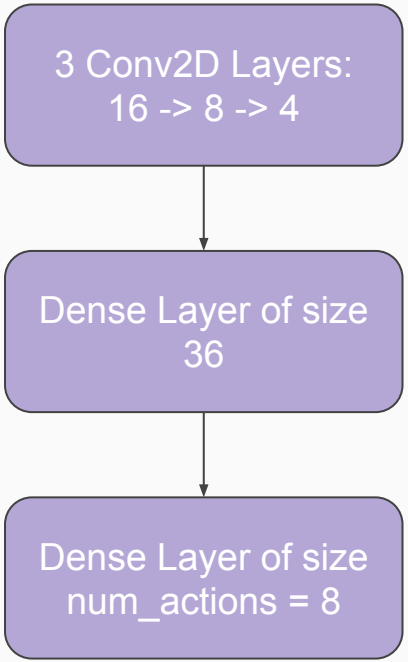
Architectures:

- Reinforce
- A2C
- DQN with Our Model

Baselines:

- Random Bot
- Smart Bot (uses the compass information, important)

# Our Model:
*DQN+*




- Adam Optimizer with learning rate 0.001
- Loss Function:

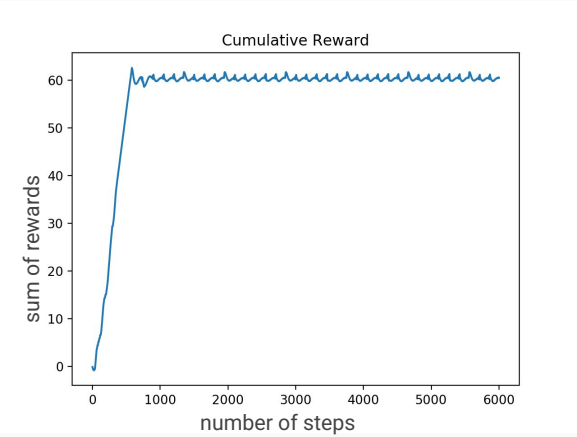
$$L = Q(s,a) - (R(s,a,s') + \gamma(max_a Q(s,a))$$

  This way we can discount future rewards.

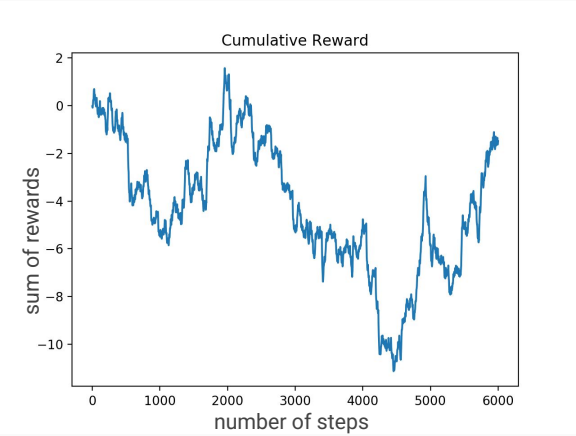
- Attempt to incorporate screen data into the Q learning more effectively with Conv2D layers by recognizing key blocks within screen

https://drive.google.com/file/d/16UOlR-XlYbiK849ulNR1HQGF1GuWtcbT/view

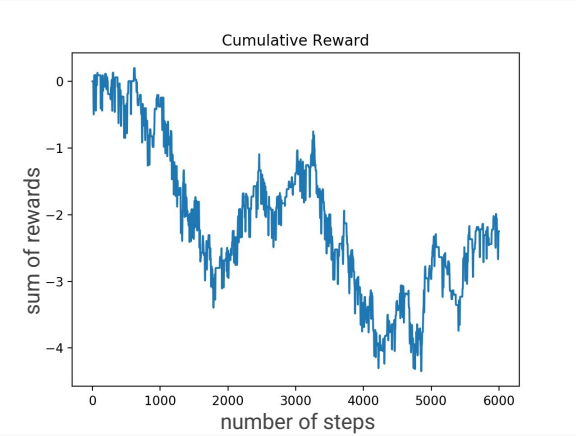
# Results:
## *Navigate Dense*




Smart Bot

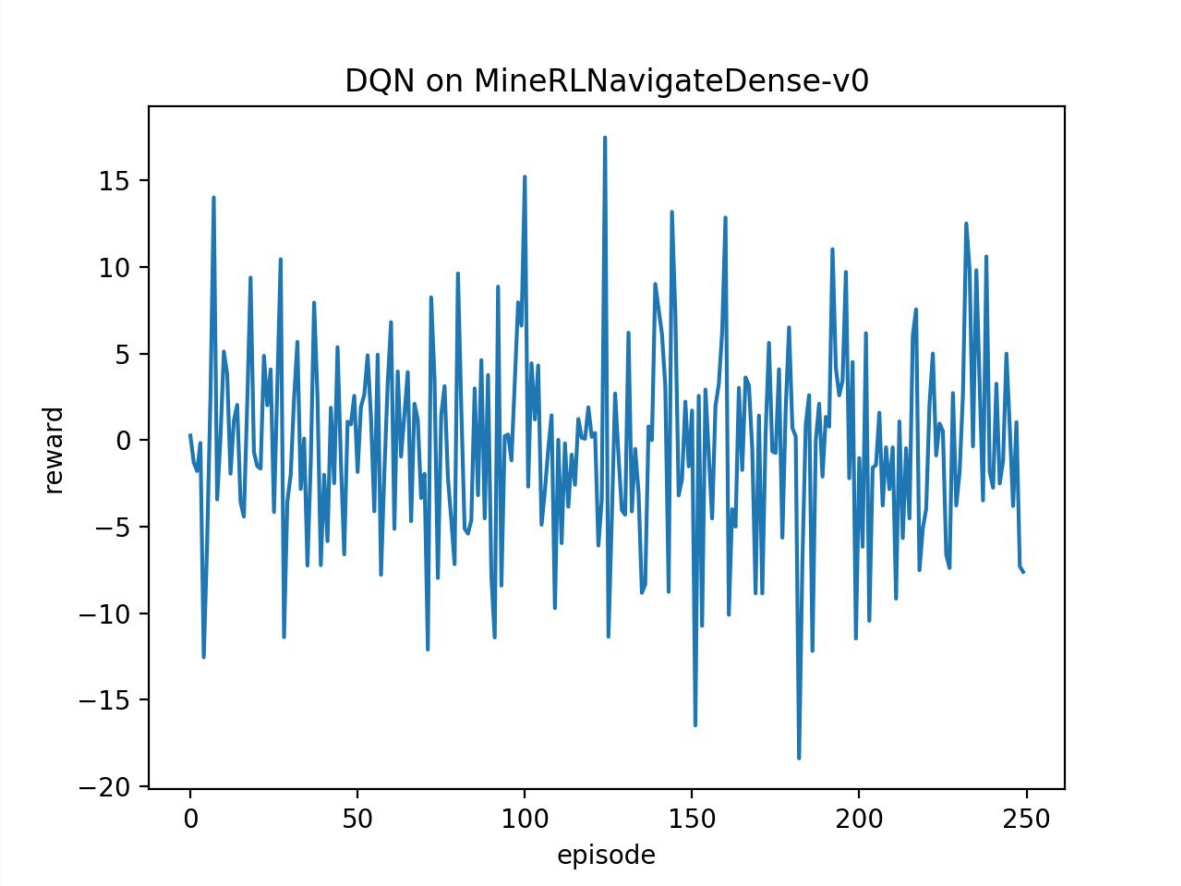



Random Bot

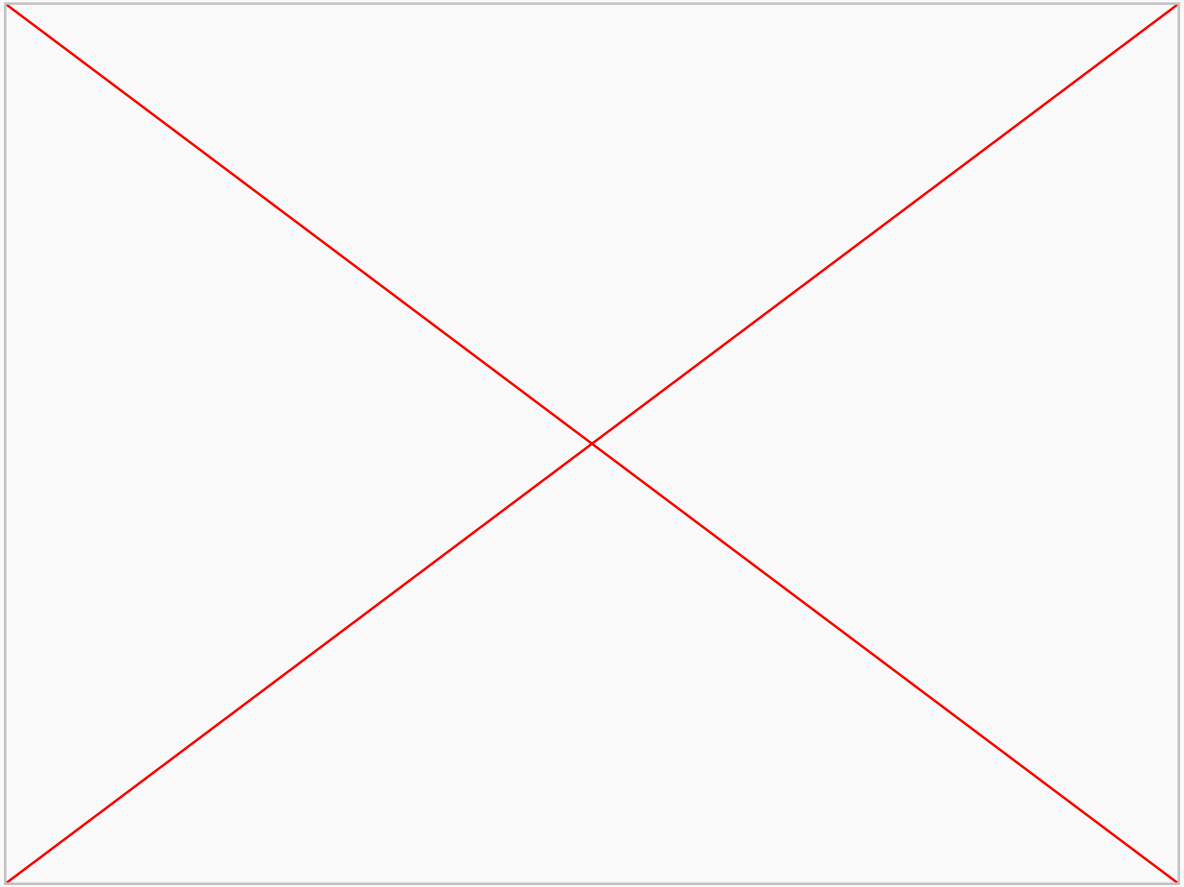



REINFORCE Bot

# Our DQN+


DQN on MineRLNavigateDense-v0
reward
episode
15
10
5
0
−5
−10
−15
−20
0
50
100
150
200
250

# Visualization: Our DQN+ in Action



# Discussion:
## *Navigate Dense*



**Challenges:**

- A fairly large action space.
- Even if it's "dense", there are limited rewards calculating expected value of actions takes an excruciatingly long time, in which the agent can pick up bad habits.
- GCP would not accept



**Solutions:**

- We simplified the action space and did not observe a major improvement (fix camera angles, ignore the inventory). It did improve behavior but not our overall goal
- We limited the number of steps for one episode and that did not show any improvements either.
- We tried to use a docker over GCP that did not work. Our local GPU access were we could sudo to download the MineRL requirements had only 2 GPUs.





# Results + Discussion:
## *Tree-Chop*

- Tree-chop returned 0 sum of rewards for a Random Bot, our model and Reinforce & Reinforce + Baseline that was left running overnight.
- This is a good example that shows that even fairly sophisticated models won't learn a lot in a sparse environment. Even if it was supposed to get a reward from chopping a block of wood, getting to that point takes a very long time.

  => this requires an insane amount of compute that we didn't have access to given the environment

# Future Work:

*The Mine RL Human Eval Dataset*

These type of problems require “some” human engineering. If this data is widely available it makes sense to use the available resources for

While current methods in RL such as Option Learning are trying to solve this with 0 human in the loop, current state-of-the-art relies on some human data.

For future work we would like to try behavioral cloning, option learning

*“ We collected human observations from a set of four main task families, each of which we explain more about below. Throughout all tasks, the agent has access to the same set of actions and observations as a human player. All tasks have a time limit, which is a part of the observation. With the exception of “Navigate,” all tasks center around obtaining specific items and have sparse rewards (+1 only for obtaining the required items). In order of what we think the difficulty is (easy to hard). ”*

- *http://minerl.io/dataset/*

# Conclusion

Even sophisticated deep learning models have a hard time learning intelligent behavior due to this sparse reward problem.

While research on more intelligent behavior such as skill learning might be helpful to solve this problem, currently relying on some human annotation doesn’t seem to be harmful.

# Thank you for listening!

*References:*

- MineRL Competition Website: *http://minerl.io/index.html*
- Human Demonstration Dataset: *http://minerl.io/dataset/*